MATH 201 Your name: _____

Learning target 2.7, version 1

Compute derivatives of implicitly-defined functions.

Consider the curve defined implicitly by the equation

$$y^3 - x \cdot y = -6.$$

1. Use implicit differentiation to find an expression for $\dfrac{dy}{dx}$ that depends on x and y. Show all of your work and thinking clearly, using correct notation.

2. The point $(7,2)$ lies on this curve. (I promise this is true, but how would you check?)
   Find the exact slope of the tangent line to this curve at the point $(7,2)$. Show clearly how you determined this value, and label your work with proper notation.

3. (Also LT 1.8!) Figure out the equation of the tangent line $L(x)$ to this curve at the point $(7,2)$, and use your tangent line equation to make a reasonable guess about the y-coordinate of the point $(6.8, ____)$.